



NORTH SYDNEY BOYS HIGH SCHOOL

Ext 1

2018 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Ms Ziazaris

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	Total	Total
Mark	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	$\overline{100}$

Section I

10 Marks

Attempt Questions 1 – 10

Use the multiple choice answer sheet for Questions 1 – 10

- 1 When the polynomial $P(x)$ is divided by $(x + 1)(x - 3)$ the remainder is $2x + 7$.

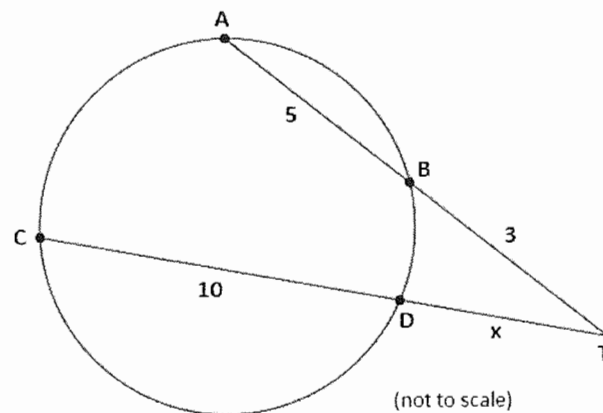
The remainder when $P(x)$ is divided by $x - 3$ is equal to

1

- (A) 1 (B) 7 (C) 9 (D) 13

- 2 In the circle below, $CD = 10$, $AB = 5$, $BT = 3$, and $DT = x$.

1



The value of x is

- (A) $x = 1 \cdot 5$ (B) $x = 2$ (C) $x = 6$ (D) $x = 4$
- 3 Which of the following is equivalent to $\frac{\sin 2\theta + \sin \theta}{\cos 2\theta + \cos \theta + 1}$?

1

- (A) $\cot \theta$ (B) $\sec \theta$ (C) $\sin \theta$ (D) $\tan \theta$

- 4 The acute angle between the lines $y = -x - 1$ and $4x + 2y = 1$ is

1

- (A) $71^\circ 34'$
(B) 45°
(C) $30^\circ 58'$
(D) $18^\circ 26'$

- 5 Given n is an integer, the general solution of $\tan\left(2x + \frac{\pi}{4}\right) = \sqrt{3}$ is 1
- (A) $x = \frac{(12n+1)\pi}{24}$
- (B) $x = \frac{(3n+1)\pi}{6}$
- (C) $x = \frac{(12n-1)\pi}{24}$
- (D) $x = \frac{(6n+1)\pi}{6}$
- 6 The solution of $x - \cos 2x = 0$ is known to be approximately 0.5 . Using one application of Newton's Method, a second approximation to this solution is 1
- (A) $x = 0.485$ (B) $x = 0.515$ (C) $x = 0.983$ (D) $x = 0.441$
- 7 A particle moves in a straight line, and its position at time t is given by $x = 3 \cos 2t + 4 \sin 2t$. Its greatest speed is: 1
- (A) 6 (B) 10 (C) 12 (D) 5
- 8 The interval AB joining the points $A(1,3)$ and $B(a,b)$ is divided internally in the ratio $2:3$ by the point $(3,13)$. The values of a and b are 1
- (A) $a = 6, b = 28$
- (B) $a = 6, b = 37$
- (C) $a = 9, b = 28$
- (D) $a = 9, b = 37$
- 9 If α, β, γ are the roots of the cubic equation $2x^3 - 5x^2 - 3x + 1 = 0$, then the value of $\alpha^2 + \beta^2 + \gamma^2$ is 1
- (A) $\frac{13}{4}$ (B) $\frac{37}{4}$ (C) $\frac{31}{4}$ (D) $\frac{19}{4}$
- 10 Which of the following is the domain of the function $y = \log_e[x + \sqrt{x^2 + 1}]$? 1
- (A) $\{x: x \leq -1, x \geq 1\}$ (B) $\{x: -1 \leq x \leq 1\}$
- (C) $\{x: x \text{ is a real number}\}$ (D) $\{x: x \geq 1\}$
-

Section II

60 Marks

Attempt Questions 11-14

Answer each question on a NEW page. Extra writing booklets are available.

In Questions 11 – 14, your responses should include all relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Start a NEW page

- (a) Find $\int \sin^2 4x \, dx$ 2
- (b) Evaluate the limit $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{6x^2}$ 2
- (c) Use the substitution $u = x^2 + 1$ to evaluate $\int_0^2 \frac{x}{(x^2+1)^3} \, dx$ 3
- (d) Find the exact value of $\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$ 3
- (e) Differentiate $y = \tan^{-1} \frac{x}{4}$, giving your answer in fully simplified form. 2
- (f) Solve the equation $\sqrt{3}\sin\theta = 1 + \cos\theta$ in the domain $0 \leq \theta \leq 2\pi$. 3

Question 12 (15 marks)

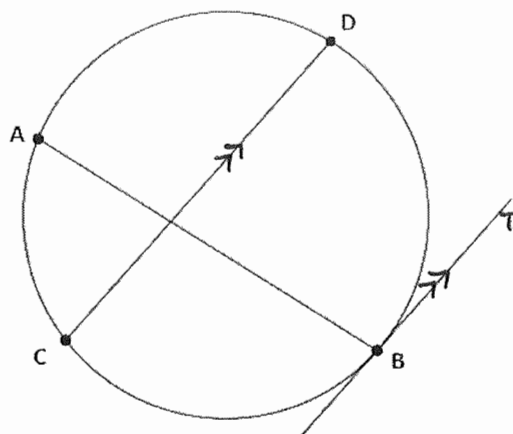
Start a NEW page

- (a) The velocity v in cm/sec of a particle moving on the x -axis is given by
 $v^2 = -3x^2 + 20x + 7$. (x is the displacement from the origin, in centimetres).

Show that the particle is moving in simple harmonic motion, and find the centre, amplitude and period of that motion.

4

- (b) AB and CD are two intersecting chords of a circle, and CD is parallel to the tangent to the circle at B .

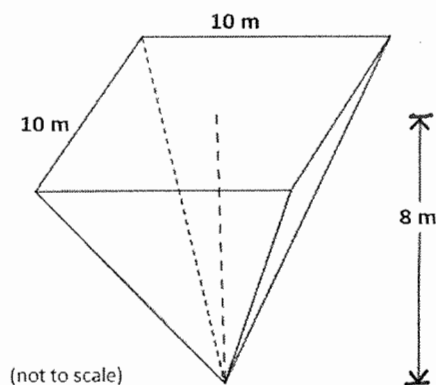


- (i) Copy or trace the diagram to your writing booklet.
 (ii) Prove that AB bisects $\angle CAD$.

3

- (c) A loading chute is in the shape of an inverted square pyramid, with base length 10 metres and depth 8 metres.

Liquid is poured in the top at a constant rate of $4 \text{ m}^3/\text{min}$.



- (i) Show that the volume V of the liquid when the level is h metres high is
 given by $V = \frac{25h^3}{48}$. [You may need to draw a diagram] 2
 (ii) At what rate is the level of the liquid rising when its depth is 4m? 2

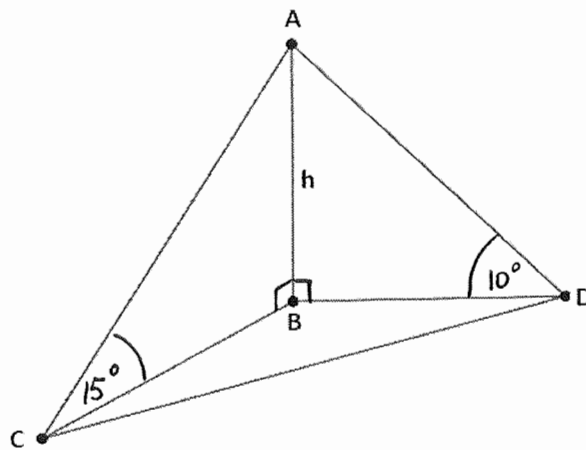
(d) For the function $y = \sin (\cos^{-1} x)$,

- | | | |
|-------|---|---|
| (i) | State the domain. | 1 |
| (ii) | State the range. | 1 |
| (iii) | Sketch the function neatly, using at least 1/3 of a page. | 2 |

Question 13 (15 marks)

Start a NEW page

(a) AB represents a communications mast, height h metres. From points C and D in the same plane as the base of the mast, the angles of elevation of the top of the mast are 15° and 10° respectively. From the base of the mast, the bearings of points C and D are 230°T and 100°T respectively.



- | | | |
|-------|--|---|
| (i) | Copy the diagram to your answer page. | |
| (ii) | Find the size of $\angle CBD$. | 1 |
| (iii) | Show that $BD = h \cot 10^\circ$. | 1 |
| (iv) | If CD is 450 metres, find the height of the mast (to 1 decimal place). | 3 |

(b) A particle P moves in a straight line with acceleration given by $\ddot{x} = 9(x - 2)$, where x is the displacement in metres from the origin O after t seconds. Initially P is 4 metres to the right of O , with velocity $v = -6$ metres per second.

- | | | |
|------|-------------------------------|---|
| (i) | Show that $v^2 = 9(x - 2)^2$ | 2 |
| (ii) | Find x as a function of t | 3 |

(c) A forensic scientist is called to a murder scene. At 10:23pm, she measures the temperature of the body as $26 \cdot 7^{\circ}\text{C}$. The ambient temperature of the room is 20°C . Exactly 60 minutes later, a second reading gives the body's temperature as $25 \cdot 8^{\circ}\text{C}$.

Assume that the victim's body before death was a normal 37°C , and that Newton's Law of cooling applies, that is, $\frac{dT}{dt} = -k(T - 20)$ where k is a constant, t is measured in minutes and temperature is in degrees Celsius,

- (i) Show that $T = 20 + Ae^{-kt}$ is a solution of the above equation. 1
- (ii) Evaluate A , then determine k to 6 decimal places. 2
- (iii) Calculate at what time of day the victim was murdered. 2

Examination continues on next page ->

Question 14 (15 marks)

Start a NEW page

- (a) Given that x is a positive integer, prove by the method of mathematical induction that $(1 + x)^n - 1$ is divisible by x for all positive integers $n \geq 1$. 3

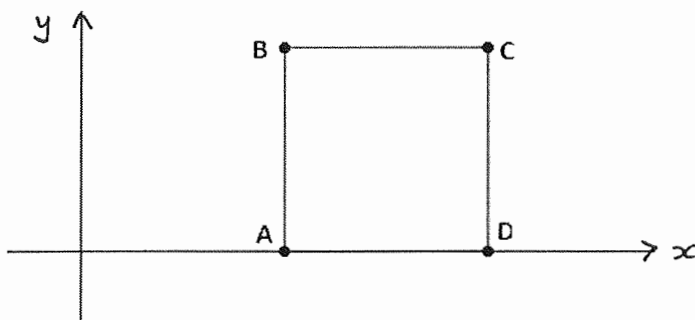
- (b) A parabola has parametric equations $x = t, y = t^2$

(i) Sketch the parabola, marking the points P and Q that correspond to $t = -1$ and $t = 2$ respectively. 1

(ii) Derive the tangents at P and Q and show they intersect at $R(\frac{1}{2}, -2)$. 2

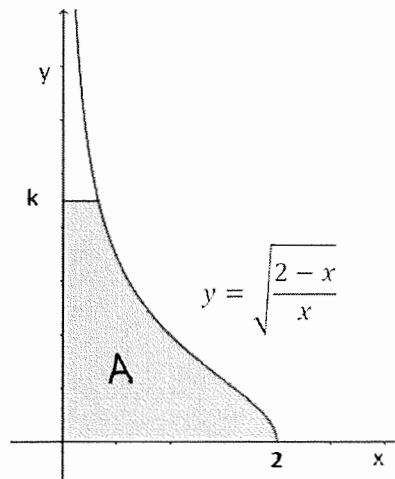
(iii) Let $T(t, t^2)$ be the point on the parabola between P and Q such that the tangent at T meets QR at the midpoint of QR . Show that the tangent at T is parallel to PQ . 3

- (c) The diagram shows a unit square with vertices $A(1, 0), B(1, 1), C(2, 1), D(2, 0)$.



A line passing through the origin with gradient m cuts the sides AB and CD at P and Q respectively. For what values of m does the line divide the area of the square in the ratio 2 : 1 ? 3

- (d) (i) Find the shaded area A . 2



- (ii) Find $\lim_{k \rightarrow \infty} A$ 1

END OF EXAMINATION

$$\textcircled{1} \quad P(x) = (x+1)(x-3) \cdot Q(x) + 2x+7$$

$$\therefore \text{rem.} = P(3) = 2(3) + 7 = 13$$

 $\therefore \textcircled{D}$

$$\textcircled{2} \quad 8(3) = (10+x)(x)$$

$$24 = x^2 + 10x$$

$$x^2 + 10x - 24 = 0$$

$$(x+12)(x-2) = 0 \quad \therefore x = 2 \quad \therefore \textcircled{B}$$

$$\textcircled{3} \quad \frac{\sin 2\theta + \sin \theta}{\cos 2\theta + \cos \theta + 1} = \frac{2 \sin \theta \cos \theta + \sin \theta}{2 \cos^2 \theta - 1 + \cos \theta + 1}$$

$$= \frac{\sin \theta (2 \cancel{\cos \theta} + 1)}{\cos \theta (2 \cancel{\cos \theta} + 1)}$$

$$= \tan \theta$$

 $\therefore \textcircled{D}$

$$\textcircled{4} \quad \tan \theta = \left| \frac{-1 - (-2)}{1 + (-1)(-2)} \right| = \left| \frac{1}{3} \right|$$

$$\theta = 18^\circ 26'$$

 $\therefore \textcircled{D}$

$$\textcircled{5} \quad \tan \left(2x + \frac{\pi}{4} \right) = \sqrt{3}$$

$$\therefore 2x + \frac{\pi}{4} = \frac{\pi}{3} + n\pi$$

$$2x = \frac{\pi}{12} + n\pi$$

$$2x = \frac{\pi + 12n\pi}{12}$$

$$= \frac{\pi(1+12n)}{24}$$

 $\therefore \textcircled{A}$

$$(6) \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(x) = x - \cos 2x$$

$$f'(x) = 1 + 2\sin 2x$$

$$\therefore x_2 = 0.5 - \frac{0.5 - \cos 1}{1 + 2\sin 1}$$

$$\approx 0.515021 \dots$$

$\therefore$ (B)

$$(7) \quad x = 3 \cos 2t + 4 \sin 2t$$

$$= 5 \cos(2t - \alpha)$$

$$\dot{x} = -10 \sin(2t - \alpha)$$

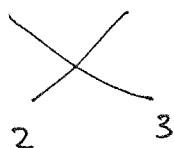
$$\therefore \text{max. speed} = 10$$

$\therefore$ (B)

(8)

A(1, 3)

B(a, b)



$$\therefore 3 = \frac{3+2a}{5} \quad \& \quad 13 = \frac{9+2b}{5}$$

$$\therefore 2a = 15 - 3 = 12$$

$$a = 6$$

$$65 = 9 + 2b$$

$$2b = 56$$

$$b = 28$$

$\therefore$ (A)

$$(9) \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= \left(\frac{5}{2}\right)^2 - 2\left(-\frac{3}{2}\right)$$

$$= \frac{25}{4} + 3 = \frac{37}{4}$$

$\therefore$ (B)

$$(10) \quad D: \text{ of } y = \ln [x + \sqrt{x^2 + 1}]$$

is all real $\mathbb{R}$

$\therefore$ (C)



$$(a) \int \sin^2 4x \, dx = \int \left(\frac{1}{2} - \frac{1}{2} \cos 8x \right) dx \quad \checkmark$$
$$= \frac{1}{2}x - \frac{1}{16} \sin 8x + C \quad \checkmark$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{6x^2}$$
$$= \lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{6x^2} \quad \checkmark$$
$$= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{6x^2}$$
$$= \frac{1}{3} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 = \frac{1}{3} \quad \checkmark$$

$$(c) I = \int_0^2 \frac{x}{(x^2+1)^3} dx$$
$$\left\{ \begin{array}{l} u = x^2 + 1 \\ du = 2x \, dx \\ x=0 \rightarrow u=1 \\ x=2 \rightarrow u=5 \end{array} \right. \quad \checkmark$$
$$= \frac{1}{2} \int_1^5 u^{-3} du$$
$$= \frac{1}{2} \left[\frac{u^{-2}}{-2} \right]_1^5 \quad \checkmark$$
$$= -\frac{1}{4} \left[\frac{1}{5^2} - \frac{1}{1^2} \right] = \frac{1}{4} \left[1 - \frac{1}{25} \right]$$
$$= \frac{1}{4} \cdot \frac{24}{25} = \frac{6}{25} \quad \checkmark$$

11 d

Then

$$\text{Let } \alpha = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right), \quad \beta = \sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$$

$$\sin \left[\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) \right]$$

$$= \sin(\alpha + \beta)$$

$$= \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

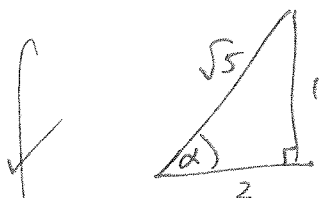
$$= \frac{1}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} + \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{10}} \quad \checkmark$$

$$= \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}}$$

$$\text{But } \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

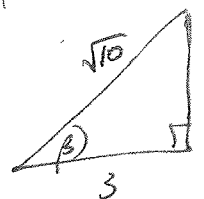
$$\therefore \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{10}}\right) = \frac{\pi}{4} \quad \checkmark$$

(Needs working for marks!)



$$\text{Let } \alpha = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$$\text{Let } \beta = \sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$$



$$(e) \quad \frac{d}{dx} \left(\tan^{-1} \frac{x}{4} \right) = \frac{1}{4} \cdot \frac{1}{1 + \left(\frac{x}{4}\right)^2} \quad \checkmark$$

$$= \frac{1}{4} \cdot \frac{1}{1 + \frac{x^2}{16}}$$

$$= \frac{1}{4} \cdot \frac{1}{\frac{16+x^2}{16}} = \frac{1}{4} \cdot \frac{16}{16+x^2}$$

$$= \frac{4}{16+x^2} \quad \checkmark$$

$$(f) \quad \sqrt{3} \sin \theta - 1 \cos \theta = 1$$

$$\text{Let } \sqrt{3} \sin \theta - 1 \cos \theta = R \sin(\theta - \alpha)$$

$$= R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$$

$$\therefore \begin{cases} R \cos \alpha = \sqrt{3} \\ R \sin \alpha = 1 \end{cases}$$

$$\therefore 2 \sin\left(\theta - \frac{\pi}{6}\right) = 1 \quad \checkmark$$

$$\sin\left(\theta - \frac{\pi}{6}\right) = \frac{1}{2}, \quad -\frac{\pi}{6} \leq \theta - \frac{\pi}{6} \leq \frac{7\pi}{6}$$

$$\therefore \theta - \frac{\pi}{6} = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

$$\therefore \theta = \frac{\pi}{3}, \pi \quad \checkmark \quad \checkmark$$

$$\begin{cases} \therefore \tan \alpha = \frac{1}{\sqrt{3}}, \\ \alpha = \frac{\pi}{6}. \\ \text{Also, } R = \sqrt{(\sqrt{3})^2 + 1^2} = 2 \end{cases}$$

11

(8)

2018

Y12 Ext.1

Task 3

Suggested Solutions

alt. let $t = \tan \frac{\theta}{2} \therefore \sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$

Thus $\sqrt{3} \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} = 1$ ✓

$$2\sqrt{3}t - 1 + \cancel{t^2} = 1 + \cancel{t^2}$$

$$2\sqrt{3}t = 2$$

$$t = \frac{1}{\sqrt{3}}$$

$$\therefore \tan \frac{\theta}{2} = \frac{1}{\sqrt{3}}, \quad 0 \leq \frac{\theta}{2} \leq \pi$$

$$\frac{\theta}{2} = \frac{\pi}{6} \therefore \theta = \frac{\pi}{3}$$
 ✓

But need to check $\theta = \pi$

$$\text{LHS} = \sqrt{3} \sin \pi - \cos \pi$$

$$= \sqrt{3}(0) - (-1)$$

$$= 1 = \text{RHS}$$

$$\therefore \theta = \pi \text{ is a soln.}$$

$$\therefore \underline{\theta = \frac{\pi}{3}, \pi}$$
 ✓

$$v^2 = -3x^2 + 20x + 7$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} \left(-\frac{3}{2} x^2 + 10x + \frac{7}{2} \right) = -3x + 10$$

$$\text{So } x = -3\left(x - \frac{10}{3}\right)$$

$$\therefore x = -(\sqrt{3})^2 \left(x - \frac{10}{3}\right)$$

which is of form $\ddot{x} = -n^2(x-x_0)$

$\therefore$ motion is SHM.

Centre of motion is $x = \frac{10}{3}$. ✓

period $T = \frac{2\pi}{n} = \frac{2\pi}{\sqrt{3}} \text{ secs.} (= \frac{2\sqrt{3}\pi}{3} \text{ secs.}) \checkmark$

At endpts, $v=0 \quad \therefore 3x^2 - 20x - 7 = 0$
 $(3x + 1)(x - 7) = 0$
 $x = -\frac{1}{3}, 7$ are endpts.

$$\therefore \text{amplitude} = \frac{1}{2} \left(\frac{1}{3} + 7 \right) = \frac{1}{2} \left(\frac{22}{3} \right) = \frac{11}{3} \text{ cm} \quad \checkmark$$

(i)

Then <

Also,

4)

(ii) Join the vertices A, B, C, D.

Then $\angle DBT = \angle BAD$ (alternate segment theorem) ✓

Also, $\angle DBT = \angle CDB$ (alternate angles,
 $CD \parallel BT$) ✓

But $\angle CDB = \angle CAB$ (angles in same segment) ✓

$$\therefore \angle BAD = \angle CAB$$

ie AB bisects $\angle CAD$. #

Q12

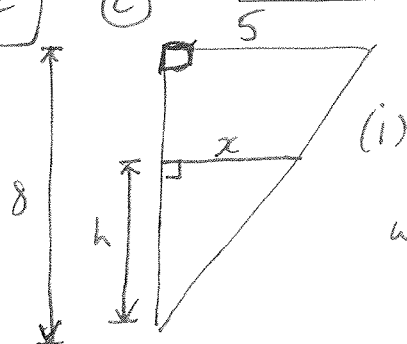
2018

Y12 Ext.1

Task 3

Suggested Solutions

(c)



$$\therefore \frac{x}{h} = \frac{5}{8} \text{ by similarity. } \checkmark$$

when height = h,

$$V = \frac{1}{3} (2x)^2 \cdot h$$

$$\text{But } x = \frac{5h}{8} \therefore 2x = \frac{5h}{4}$$

$$\therefore V = \frac{1}{3} \cdot \left(\frac{5h}{4}\right)^2 \cdot h = \frac{25h^3}{48} \quad \checkmark$$

$$(ii) \quad \frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$\therefore \frac{dV}{dt} = \frac{25h^2}{16} \cdot \frac{dh}{dt} \quad \checkmark$$

$$\therefore 4 = \frac{25(4^2)}{16} \cdot \frac{dh}{dt}$$

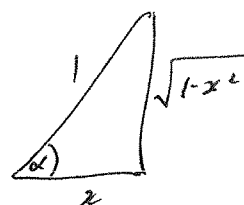
$$\therefore \left(\frac{dh}{dt} = \frac{4}{25} \text{ m/min.} \right. \\ \left. (= 0.16 \text{ m/min}) \right) \quad \checkmark$$

$$(d) \quad y = \sin(\cos^{-1} x)$$

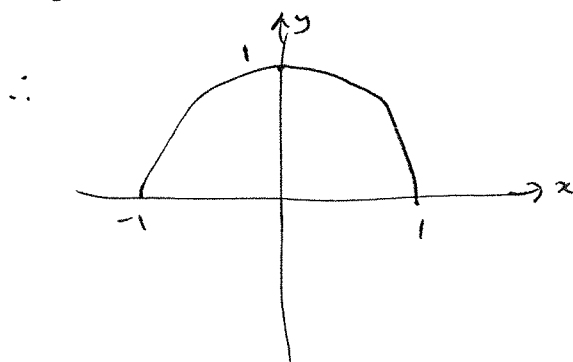
$$(i) \quad D: -1 \leq x \leq 1 \quad \checkmark$$

$$(ii) \quad R: 0 \leq y \leq 1 \quad \checkmark$$

$$(iii) \quad \text{Let } \alpha = \cos^{-1} x \therefore x = \cos \alpha$$



$$\therefore y = \sin \alpha = \sqrt{1-x^2}$$



✓✓

13

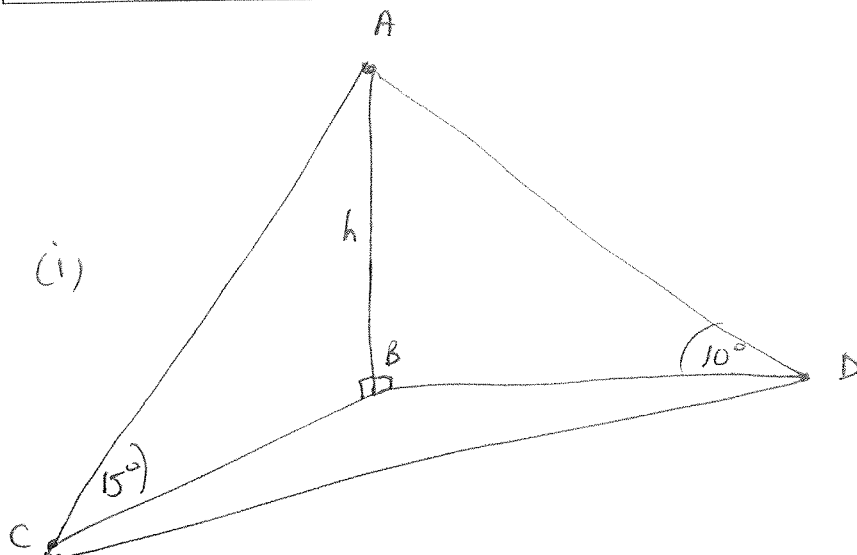
2018

Y12 Ext.1

Task 3

Suggested Solutions

(a)

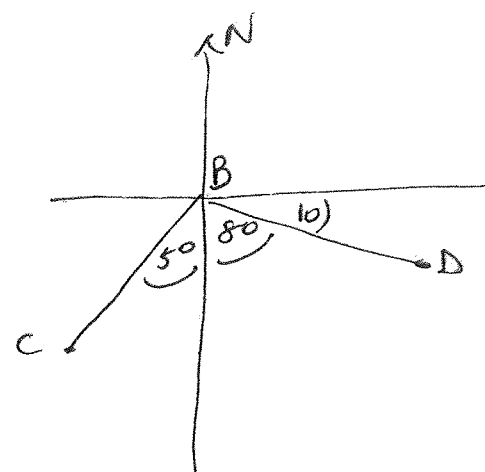


(ii) $\angle CBD = 130^\circ$ ✓

(iii) $\frac{h}{BD} = \tan 10^\circ$ ✓

$$BD = \frac{h}{\tan 10^\circ}$$

$$\therefore BD = h \cot 10^\circ$$



(iv) by cosine rule in $\triangle CBD$,

$$(CD = h \cot 15^\circ)$$

$$450^2 = (h \cot 10^\circ)^2 + (h \cot 15^\circ)^2 - 2 \cdot h \cot 10^\circ \cdot h \cot 15^\circ \cdot \cos 130^\circ$$

$$= h^2 [\cot^2 10^\circ + \cot^2 15^\circ - 2 \cot 10^\circ \cdot \cot 15^\circ \cdot \cos 130^\circ]$$

$$\therefore h^2 = \frac{450^2}{\cot^2 10^\circ + \cot^2 15^\circ - 2 \cot 10^\circ \cdot \cot 15^\circ \cdot \cos 130^\circ}$$

$$\doteq 2762.5629 \dots$$

$$\therefore h \doteq 52.5600$$

$$\therefore h = 52.6 \text{ m (1 d.p.)}$$

13 (b) $\ddot{x} = 9/(x-2)$

at $t=0$, $x=+4$, $v=-6$.

(i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 9x - 18$

$$\frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + C$$

at $t=0$, $x=4$, $v=-6$

$$\therefore \frac{1}{2} \cdot 36 = \frac{9}{2} \cdot 16 - 18 \cdot 4 + C \quad \therefore C = 18$$

$$\therefore \frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + 18$$

$$v^2 = 9x^2 - 36x + 36$$

$$= 9(x^2 - 4x + 4)$$

$$\therefore v^2 = 9(x-2)^2$$

(ii) $v = \pm 3(x-2)$ But: when $x=+4$, $v=-6$
 $\therefore v = -3(x-2)$

$$\therefore \frac{dx}{dt} = -3(x-2) \quad \therefore \frac{dt}{dx} = -\frac{1}{3} \cdot \frac{1}{x-2}$$

$$\therefore t = -\frac{1}{3} \ln(x-2) + C$$

at $t=0$, $x=4$

$$\therefore 0 = -\frac{1}{3} \ln(2) + C$$

$$\therefore C = \frac{1}{3} \ln 2$$

$$\therefore t = -\frac{1}{3} \ln(x-2) + \frac{1}{3} \ln 2$$

$$= \frac{1}{3} (\ln 2 - \ln(x-2)) = \frac{1}{3} \ln \left(\frac{2}{x-2} \right)$$

$$\therefore e^{3t} = \frac{2}{x-2}$$

$$\therefore x-2 = 2e^{-3t}$$

$$\therefore x = 2 + 2e^{-3t}$$

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(c) we're given $\frac{dT}{dt} = -k(T-20)$.

(i) If $T = 20 + Ae^{-kt}$, i.e. $Ae^{-kt} = T-20$,

$$\text{Then } \frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T-20)$$

$\therefore$ a solution.

(ii) Let $t=0$ at 10:23 pm.

at $t=0$, $T = 26.7$

$$\therefore 26.7 = 20 + Ae^0$$

$$A = 6.7$$

at $t=60$, $T = 25.8$

$$\therefore 25.8 = 20 + 6.7e^{-60k}$$

$$\therefore e^{-60k} = \frac{5.8}{6.7}$$

$$-60k = \ln\left(\frac{5.8}{6.7}\right)$$

$$\therefore k = -\frac{1}{60} \ln\left(\frac{5.8}{6.7}\right)$$

$$k = 0.002404 \text{ (6dp)}$$

(iii) we want t when $T = 37$;

$$\therefore 37 = 20 + 6.7e^{-kt}$$

$$\therefore e^{-kt} = \frac{17}{6.7} \quad \therefore -kt = \ln\left(\frac{17}{6.7}\right)$$

$$t = -\frac{1}{0.002404} \cdot \ln\left(\frac{17}{6.7}\right) = -387.315 \text{ mins.}$$

$$= -6 \text{ h, } 27.3 \text{ mins}$$

$\therefore$ 3:56 pm was time of death.

(a) When $n=1$

$$(1+x)^1 - 1 = 1+x-1 = x,$$

which is divisible by x .

$\therefore$ true for $n=1$ ✓

Assume true for $n=k$

i.e. assume $(1+x)^k - 1 = Mx$
for some integer M .

$$\text{i.e. } (1+x)^k = Mx + 1$$

Then for $n=k+1$,

$$(1+x)^{k+1} - 1 = (1+x)(1+x)^k - 1$$

$$= (1+x)(Mx+1) - 1 \quad \text{by assumption} \quad \checkmark$$

$$= (Mx+1) + (Mx^2+x) - 1$$

$$= (M+1)x + Mx^2$$

$$= x(M+1+Mx)$$

$$= Nx, \quad N \text{ an integer, as } M, x \text{ are integers.} \quad \checkmark$$

$\therefore$ true for $n=k+1$ if true for $n=k$.

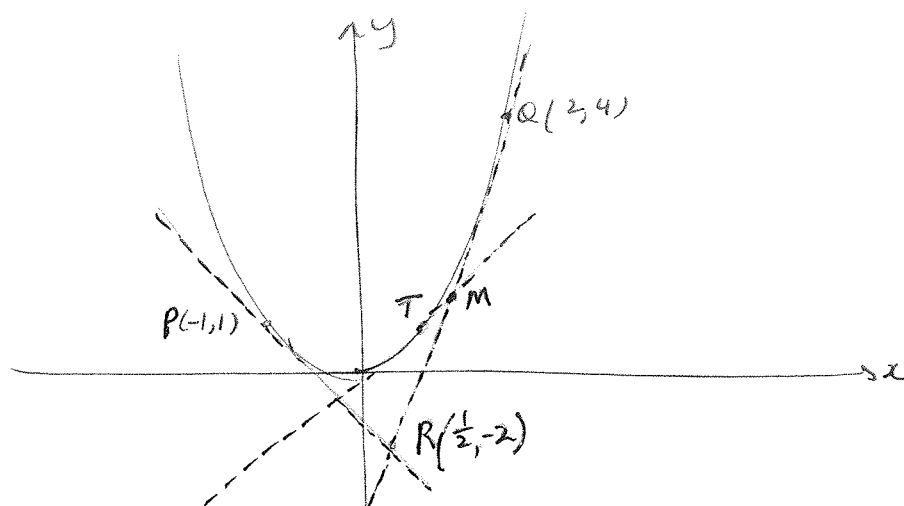
Hence true by induction. #

14 (b) $x=t, y=t^2 \therefore$ Cartesian eqn is $y=x^2$

$$t = -1 \Rightarrow x = -1, y = 1$$

$$t = 2 \Rightarrow x = 2, y = 4.$$

(i)



(ii) $y = x^2 \therefore \frac{dy}{dx} = 2x$

$$\therefore \text{tangent at P is } y - 1 = -2(x + 1)$$

$$\text{Likewise, tangent at Q is } y - 4 = 4(x - 2)$$

Solve simult. for R:

$$1 - 2(x + 1) = 4 + 4(x - 2)$$

$$1 - 2x - 2 = 4 + 4x - 8$$

$$\therefore 6x = 3 \quad \therefore x = \frac{1}{2}$$

$$\text{Thus } y = 1 - 2\left(\frac{1}{2} + 1\right) = 1 - 3 = -2$$

$$\therefore R = \left(\frac{1}{2}, -2\right)$$

(iii)

Tangent at T is

$$y - t^2 = 2t(x - t)$$

$$M = \text{Midpt of } QR = \left(\frac{2 + \frac{1}{2}}{2}, \frac{4 - 2}{2}\right) = \left(\frac{5}{4}, 1\right) \checkmark$$

M satisfies tangent at T

$$\therefore 1 - t^2 = 2t\left(\frac{5}{4} - t\right)$$

$$4 - 4t^2 = 10t - 8t^2$$

$$4t^2 - 10t + 4 = 0$$

$$2t^2 - 5t + 2 = 0, \quad (2t - 1)(t - 2) = 0$$

$$\therefore t = \frac{1}{2} \text{ or } t = 2$$

$\rightarrow$

(b) (ii) etc.

But $t \neq 2$ (as $t = 2$ at Q)

$$\therefore t = \frac{1}{2} \text{ at } T$$

tangent at is $y - t^2 = 2t(x - t)$

$$\therefore M_T \text{ gradient is } 2t = 2\left(\frac{1}{2}\right) = 1.$$

$$M_{PQ} = \frac{4-1}{2+1} = 1$$

$$\therefore M_T = M_{PQ} \text{ as } \text{negative.}$$

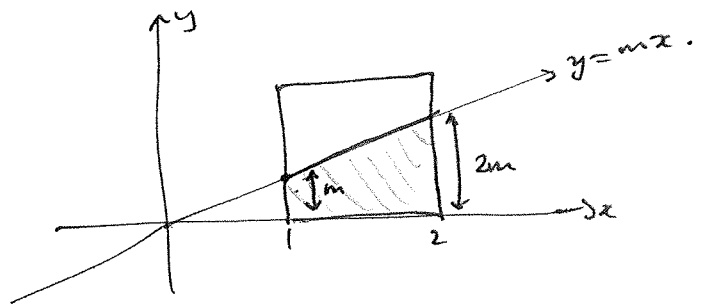
 $\therefore \text{parallel.} \quad \#$


(c)

Let the line be $y = mx$.

$$\therefore \text{when } x = 1, \quad y = m$$

$$\text{when } x = 2, \quad y = 2m.$$



Thus shaded area is

$$A = \frac{m+2m}{2} \times 1 = \frac{3m}{2}, \quad \checkmark$$

$$\therefore \text{remaining area is } 1 - \frac{3m}{2} = \frac{2-3m}{2}$$

Thus we have

$$\frac{\frac{3m}{2}}{\frac{2-3m}{2}} = \frac{2}{1} \quad \text{or} \quad \frac{\frac{3m}{2}}{\frac{2-3m}{2}} = \frac{1}{2} \quad \checkmark$$

$$\therefore 3m = 4 - 6m$$

$$6m = 4 - 6m$$

$$m = \frac{4}{9} \quad \text{or} \quad m = \frac{2}{9} \quad \checkmark$$

(d)

(i) We integrate along the y -axis:Now, if $y = \sqrt{\frac{2-x}{x}}$ then

$$y^2 = \frac{2-x}{x}$$

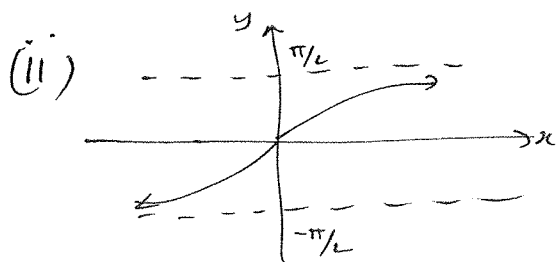
$$\therefore xy^2 = 2-x$$

$$x(y^2+1) = 2$$

$$\therefore x = \frac{2}{1+y^2} \quad \checkmark$$

$$\therefore \text{Area } A = \int_{y=0}^{y=k} \frac{2}{1+y^2} dy = 2 \left[\tan^{-1} y \right]_0^k$$

$$A = 2 \tan^{-1} k \text{ units}^2. \quad \checkmark$$



$$\therefore \text{as } k \rightarrow \infty, \quad \tan^{-1} k \rightarrow \frac{\pi}{2}$$

$$\therefore A \rightarrow \pi$$

$$\text{So } \lim_{k \rightarrow \infty} A = \pi \text{ units}^2. \quad \checkmark$$

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